# A SHARP HARDY INEQUALITY ON $\mathbb{Z}^2$

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November 4, 2026

Abstract. We prove the Hardy inequality on $\mathbb{Z}^2$ with Euclidean weight $|x|^{-2}(\log|x|)^{-2}$ and sharp constant $1/4$, for finitely supported functions vanishing on $\{|x|\le 1\}$.

## 1. Introduction

On the exterior domain $\Omega=\{x\in\mathbb{R}^2:|x|>1\}$, the classical logarithmic Hardy inequality

$$\int_\Omega |\nabla u(x)|^2\,dx \ge \frac{1}{4}\int_\Omega \frac{|u(x)|^2}{|x|^2(\log|x|)^2}\,dx$$

holds for every $u\in C_c^\infty(\Omega)$. The constant $1/4$ is sharp. We establish the corresponding inequality for the nearest-neighbor Laplacian on $\mathbb{Z}^2$, with the same Euclidean weight and the same sharp constant, for finitely supported functions vanishing on $\{|x|\le 1\}$.

Throughout, $|x|$ denotes the Euclidean norm and log the natural logarithm. Let $e_1=(1,0)$, $e_2=(0,1)$, and set

$$K=\{x\in\mathbb{Z}^2:|x|\le 1\}=\{0,\pm e_1,\pm e_2\},\qquad D=\mathbb{Z}^2\setminus K.$$

We identify $C_c(D)$ with the real-valued finitely supported functions on $\mathbb{Z}^2$ that vanish on $K$. Their Dirichlet energy is

$$\mathcal{E}(u)=\sum_{x\in\mathbb{Z}^2}\sum_{j=1}^{2}|u(x+e_j)-u(x)|^2, \tag{1.1}$$

so that each unoriented edge is counted once.

**Theorem 1.1.** *Every $u\in C_c(D)$ satisfies*

$$\sum_{x\in\mathbb{Z}^2}\sum_{j=1}^{2}|u(x+e_j)-u(x)|^2 \ge \frac{1}{4}\sum_{x\in D}\frac{|u(x)|^2}{|x|^2(\log|x|)^2}. \tag{1.2}$$

*The constant $1/4$ is sharp, and the inequality is strict for every nonzero $u\in C_c(D)$.*

Kapitanski and Laptev [3, Theorem 4.1] established a logarithmic Hardy inequality on $\mathbb{Z}^2$ with an unspecified positive coefficient. Their theorem uses the same Dirichlet set $K$, with the Euclidean weighted sum restricted to $|x|\ge 2$. Inequality (1.2) also includes the four points $(\pm1,\pm1)$.

Huang and Ye [2, Remark 6.2] obtained a logarithmic Hardy weight on $\mathbb{Z}^2$ with leading coefficient $1/4$, for functions supported on $\{|x|\ge 2\}$. Its next term is negative along diagonal directions, so their expansion does not establish (1.2) with the prescribed weight.

Hake, Keller and Pogorzelski [1, Corollary 2 and the subsequent remark] constructed an optimal Hardy weight from the lattice potential kernel. Its leading term is $1/[4|x|^2(\log|x|)^2]$, followed by a negative correction of order $|x|^{-2}(\log|x|)^{-3}$. They proved sharpness of the leading coefficient near infinity and discussed whether $1/4$ remains valid globally for the prescribed logarithmic weight. Theorem 1.1 establishes this endpoint for $K$. Their criticality result concerns the different class of functions vanishing only at the origin.

*Key words and phrases.* Discrete Hardy inequality, square lattice, sharp constant, supersolution, logarithmic weight.

The proof uses the ground-state identity [4]. We start from the radial function $\sqrt{\frac{1}{2}\log(|x|^2-\frac{1}{2})}$. An angular comparison bounds its four-neighbor sum by the corresponding sum in a diagonal direction, where an exact factorization gives the required estimate. Changing the values on the two orbits $|x|^2=2,4$ then produces a global supersolution. The construction yields a positive remainder, while logarithmic cutoffs prove sharpness.

A direct consequence is the following family of inequalities.

**Corollary 1.2.** *For each $b\ge 1$, every $u\in C_c(D)$ satisfies the sharp inequality*

$$\mathcal{E}(u)\ge \frac{1}{4}\sum_{x\in D}\frac{|u(x)|^2}{|x|^2\log^2(b|x|)}. \tag{1.3}$$

The inequality follows from (1.2) because $\log(b|x|)\ge \log|x|$ on $D$; sharpness is proved in Section 3. The inequalities and their sharp constants also hold for complex-valued functions, by applying the inequalities to the real and imaginary parts.

## 2. The supersolution

Write $x\sim y$ when $|x-y|=1$, and define

$$L\phi(x)=4\phi(x)-\sum_{y\sim x}\phi(y),\qquad W(x)=\frac{1}{4|x|^2(\log|x|)^2}\quad (x\in D).$$

Thus $\mathcal{E}(u)=\sum_x u(x)Lu(x)$. Our goal is to construct $\phi>0$ on $D$, with $\phi=0$ on $K$, such that $L\phi>W\phi$ on $D$. The following identity explains this reduction.

**Lemma 2.1.** *If $\phi>0$ on $D$ and $\phi=0$ on $K$, then for every $u\in C_c(D)$,*

$$\mathcal{E}(u)=\sum_{\substack{\{x,y\}:x\sim y\\ x,y\in D}}\phi(x)\phi(y)\left(\frac{u(x)}{\phi(x)}-\frac{u(y)}{\phi(y)}\right)^2+\sum_{x\in D}\frac{L\phi(x)}{\phi(x)}u(x)^2. \tag{2.1}$$

*Each edge in the first sum is counted once.*

*Proof.* For $x,y\in D$, expansion gives

$$\begin{aligned}(u(x)-u(y))^2=\phi(x)\phi(y)\left(\frac{u(x)}{\phi(x)}-\frac{u(y)}{\phi(y)}\right)^2\\ +\frac{\phi(x)-\phi(y)}{\phi(x)}u(x)^2+\frac{\phi(y)-\phi(x)}{\phi(y)}u(y)^2.\end{aligned}$$

Sum over the edges in $D$. An edge from $x\in D$ to $K$ contributes $u(x)^2$ to the energy and the same amount to the potential term, since $u=\phi=0$ on $K$. This proves (2.1). □

**Lemma 2.2.** *Let $q>0$, and suppose that $F\in C^4$ on an open interval containing $[q+1-2\sqrt{q},q+1+2\sqrt{q}]$, with $F^{(4)}\le 0$ there. If $x_1^2+x_2^2=q$, then*

$$\begin{aligned}\sum_{j=1}^{2}\big(F(q+1+2x_j)+F(q+1-2x_j)\big)\\ \le 2F(q+1+\sqrt{2q})+2F(q+1-\sqrt{2q}).\end{aligned} \tag{2.2}$$

*Proof.* Put $a=q+1$ and $G(s)=F(a+2\sqrt{s})+F(a-2\sqrt{s})$ for $0\le s\le q$. For $s>0$, with $t=2\sqrt{s}$, differentiation gives

$$G''(s)=\frac{4}{t^3}\left[t\big(F''(a+t)+F''(a-t)\big)-\int_{a-t}^{a+t}F''(v)\,dv\right]\le 0.$$

Indeed, $F''$ is concave and therefore lies above the chord joining its endpoint values. By continuity, $G$ is concave on $[0,q]$. Jensen's inequality now gives $G(x_1^2)+G(x_2^2)\le 2G(q/2)$, which is (2.2). □

We use the function

$$F(v)=\sqrt{\frac{1}{2}\log\left(v-\frac{1}{2}\right)}\qquad (v>3/2)$$

and define

$$\phi(x)=\begin{cases}0, & |x|^2\le 1,\\ \dfrac{3}{5}, & |x|^2=2,\\ \dfrac{18}{25}, & |x|^2=4,\\ F(|x|^2), & |x|^2\ge 5.\end{cases}\tag{2.3}$$

These cases exhaust $\mathbb{Z}^2$, and $\phi>0$ on $D$.

**Proposition 2.3.** *The function (2.3) satisfies $(L-W)\phi>0$ on $D$. More precisely, for $q=|x|^2\ge 8$,*

$$\frac{L\phi(x)}{\phi(x)}\ge\frac{1}{q\log^2(q-\frac{1}{2})}>W(x),\tag{2.4}$$

*and for $1<|x|^2<8$, we have $(L-W)\phi(x)>1/1250$.*

*Proof.* Suppose first that $q\ge 8$. Then $\phi(x)=F(q)$ and $\phi(y)\le F(|y|^2)$ for every neighbor $y$ of $x$. Indeed, a neighbor can meet a modified orbit only when $q=9$ and $|y|^2=4$, and $18/25<F(4)$, as verified below. Writing $t=\log(v-\frac{1}{2})>0$, we have

$$F^{(4)}(v)=-\frac{48t^3+44t^2+36t+15}{16\sqrt{2}\,(v-\frac{1}{2})^4t^{7/2}}<0.$$

Since $(\sqrt{q}-1)^2\ge(\sqrt{8}-1)^2>3/2$, Lemma 2.2 applies to the four neighboring values. Set

$$h=\log\left(q-\frac{1}{2}\right),\qquad d=\log\frac{\sqrt{2q}+1}{\sqrt{2q}-1}.$$

The factorization $q+\frac{1}{2}\pm\sqrt{2q}=\frac{1}{2}(\sqrt{2q}\pm 1)^2$ gives

$$\log\left(q+\frac{1}{2}\pm\sqrt{2q}\right)=h\pm d,\qquad 0<d<h.$$

The elementary inequalities

$$\sqrt{1+z}+\sqrt{1-z}\le 2-\frac{z^2}{4},\qquad \log\frac{1+z}{1-z}\ge 2z\qquad (0\le z<1)$$

follow, respectively, by squaring and by integrating $2/(1-z^2)\ge 2$. In particular, $d\ge\sqrt{2/q}$. The angular comparison therefore gives

$$\begin{aligned}\frac{L\phi(x)}{\phi(x)}&\ge 4-2\sqrt{1+d/h}-2\sqrt{1-d/h}\\ &\ge\frac{d^2}{2h^2}\ge\frac{1}{qh^2}>\frac{1}{q(\log q)^2}=W(x).\end{aligned}$$

This proves (2.4).

It remains to check $q = 2, 4, 5$. Write $W_q = 1/[q(\log q)^2]$. We use the rational bounds

$$\frac{18}{25} < F(4), \qquad \frac{173}{200} < F(5) < \frac{87}{100},$$

$$F(8) < \frac{201}{200}, \qquad F(9) < \frac{26}{25}, \qquad F(10) < \frac{213}{200}, \tag{2.5}$$

$$W_2 < \frac{21}{20}, \qquad W_4 < \frac{2}{15}, \qquad W_5 < \frac{2}{25}. \tag{2.6}$$

All these bounds follow by substitution in the elementary enclosure

$$0 < \log t - 2\sum_{k=0}^{4} \frac{z^{2k+1}}{2k+1} < \frac{2z^{11}}{11(1-z^2)}, \qquad z = \frac{t-1}{t+1}, \quad t > 1. \tag{2.7}$$

To obtain it, expand $\log t = 2\sum_{k\geq 0} z^{2k+1}/(2k+1)$ and bound the tail by a geometric series. Substituting (2.5)–(2.6) into the three neighbor sums gives

$$\begin{aligned}
(L-W)\phi(1,1) &= (4-W_2)\frac{3}{5} - 2F(5) > \frac{3}{100},\\
(L-W)\phi(2,0) &= (4-W_4)\frac{18}{25} - F(9) - 2F(5) > \frac{1}{250},\\
(L-W)\phi(2,1) &= (4-W_5)F(5) - \frac{3}{5} - \frac{18}{25} - F(8) - F(10) > \frac{1}{1250}.
\end{aligned} \tag{2.8}$$

Sign changes and coordinate permutations cover all remaining points. □

*Proof of the lower bound in Theorem 1.1.* Apply (2.1) to the function (2.3). The first term is nonnegative, and Proposition 2.3 gives $L\phi/\phi > W$ at every point of $D$. Hence $\mathcal{E}(u) \geq \sum_D W(x)u(x)^2$, with strict inequality for $u \neq 0$. □

The same estimates give an explicit remainder.

**Corollary 2.4.** *For $x \in D$, put $q = |x|^2$ and*

$$R(x) = \begin{cases} \dfrac{1}{1250}, & 1 < q < 8,\\[2ex] \dfrac{1}{q\log^2(q-\frac{1}{2})} - \dfrac{1}{q(\log q)^2}, & q \geq 8. \end{cases}$$

*Then $R(x) > 0$ on $D$, and every $u \in C_c(D)$ satisfies*

$$\mathcal{E}(u) \geq \frac{1}{4}\sum_{x\in D} \frac{|u(x)|^2}{|x|^2(\log|x|)^2} + \sum_{x\in D} R(x)|u(x)|^2. \tag{2.9}$$

*Proof.* For $1 < q < 8$, we have $\phi(x) < 1$, so (2.8) gives $(L-W)\phi(x)/\phi(x) > 1/1250$. For $q \geq 8$, use (2.4). The conclusion follows from (2.1). □

The coefficient $1/4$ is therefore sharp within the family of constant multiples of the prescribed weight, although the weight admits a positive additive improvement. No optimality is asserted for the remainder $R$.

## 3. Sharpness

Choose $\chi \in C^\infty([0,\infty))$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[0,1]$, and $\chi = 0$ on $[2,\infty)$. For $T \geq 2$, put

$$u_T(x) = \begin{cases} \sqrt{\log|x|}\,\chi(T^{-1}\log|x|), & x \in D,\\ 0, & x \in K. \end{cases}$$

These functions are finitely supported. Write

$$\mathcal{V}_b(u) = \sum_{x\in D} \frac{|u(x)|^2}{|x|^2 \log^2(b|x|)} \quad (b \geq 1), \qquad \mathcal{V} = \mathcal{V}_1.$$

Let $U_T(z) = \sqrt{\log|z|}\,\chi(T^{-1}\log|z|)$ for $|z| > 1$. On $r = |z| \geq 2$, differentiation gives, uniformly in $T$,

$$|\nabla U_T(z)| \leq Cr^{-1}, \qquad |\nabla^2 U_T(z)| \leq Cr^{-2}.$$

The gradients of $|\nabla U_T|^2$ and $U_T^2/(r^2\log^2 r)$ are consequently bounded by $Cr^{-3}$. Taylor's theorem and comparison on unit squares therefore give uniformly bounded errors when replacing lattice sums by integrals outside a fixed ball, since $\sum_{|x|\geq 3}|x|^{-3} < \infty$. The finitely many remaining lattice terms are also uniformly bounded.

With $t = \log r$, it follows that

$$\begin{aligned}
\mathcal{V}(u_T) &= 2\pi \int_{\log 2}^{\infty} \frac{\chi(t/T)^2}{t}\,dt + O(1) = 2\pi \log T + O(1),\\
\mathcal{E}(u_T) &= 2\pi \int_{\log 2}^{\infty} \left(\frac{\chi(t/T)}{2\sqrt{t}} + \frac{\sqrt{t}}{T}\chi'(t/T)\right)^2 dt + O(1)\\
&= \frac{\pi}{2}\log T + O(1).
\end{aligned}$$

Indeed, $\chi' = 0$ unless $T \leq t \leq 2T$, so the terms containing $\chi'$ contribute $O(1)$. Hence $\mathcal{E}(u_T)/\mathcal{V}(u_T) \to 1/4$, proving sharpness in Theorem 1.1.

For fixed $b \geq 1$, the bound $u_T(x)^2 \leq \log|x|$ gives

$$0 \leq \mathcal{V}(u_T) - \mathcal{V}_b(u_T) \leq 2(\log b) \sum_{x\in D} \frac{1}{|x|^2(\log|x|)^2} = O_b(1).$$

The series converges, so $\mathcal{E}(u_T)/\mathcal{V}_b(u_T) \to 1/4$, proving Corollary 1.2. Removing any fixed finite set from the weighted sums changes them by $O(1)$; the coefficient remains sharp after such a removal.

## References


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